coming from the piston head during power stroke is observed by the slot. By this way the piston is protected from expansion and allowed to operate the engine safely.



Figure 5.3.7.1(a) Horizontal Slot

b) Heat Dam:

It is like how we store water in ponds and use it when needed. Similar to that to save the heat generated inside the engine combustion chamber and use it effectively small pits in concave shape are made on the piston head which are used to save heat. Hence the heat transfered from the crown to the skirt is being reduced and the expansion of piston is being avoided

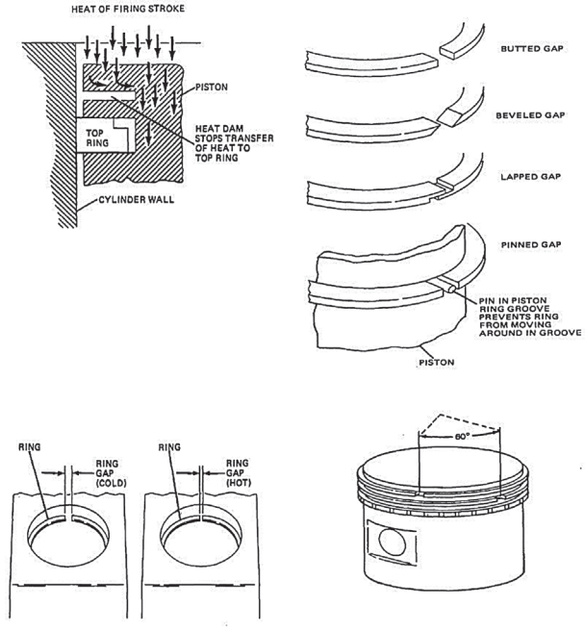


Figure 5.3.7.1(b) Heat Dam

c) Vertical Slot:

This type of slots is used in preventing the piston's expansion due to heat particularly at the outer periphery. The slots are made in selected places where the expansion is predicted to be more in such a way that it should tolerate the expansion of the piston.



Figure 5.3.7.1(c) Vertical Slot

d) T slot:

Similar to the vertical slot piston the T-slot helps the Piston to avoid expansion in the diameter wise as well as in the lengthwise and helps engine to function properly.



Figure 5.3.7.1(d) T-Slot

e) Oblique Slot:

In this type the slots are made on the piston's oil ring slots to control the temperature of the piston. The slots are made slightly in slanting manner such that it reaches the skirt. In this type the heat produced in the engine is blocked in such a way that it should not reach the skirt. The heat is transferred in sideways and also in the lengthwise so that the piston is saved from the high temperature produced inside the engine combustion chamber.

AM_Unit_05.indd 85 12/20/2019 5:26:32 PM